

Yuan Chang (Merlin)

Portfolio: [itis2010me.github](https://github.com/itis2010me)

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EDUCATION

- **University of California, San Diego** La Jolla, California
Master of Science - Computer Science and Engineering 2022 - 2024
- **University of California, Davis** Davis, California
Bachelor of Science - Computer Science, Minor in Mathematics; Graduated with Honors 2018 - 2022

EXPERIENCE

- **Amazon Web Services (AWS) - Automated Reasoning Group** Seattle, WA
Software Development Engineer II July 2026 - Present
Software Development Engineer I Feb 2025 - July 2026
 - Building a system that evaluates AI agent actions against Cedar and Dogwood policies to improve agent safety.
 - Migrated the service architecture to Firecracker microVMs, raising the compute fleet scaling ceiling 20x and cutting per-unit scale-up time from minutes to seconds.
 - Built stateful MCP servers exposing remote automated-reasoning solvers and provers (SAT, SMT, LEAN) to internal services and agents.
 - Improved availability during outages with automatic load balancing driven by deep health checks.
 - Cut recurring pages by hardening the service's handling of asynchronous AWS ECS events.
 - Mentored junior engineers and an intern who converted to full-time; ran the team's operational reviews.
- **Amazon Web Services (AWS) - High Performance Computing** Seattle, WA
Software Development Engineer I Apr 2024 - Feb 2025
 - Helped launch AWS Parallel Computing Service (PCS), a public AWS service for managed HPC clusters.
 - Delivered features spanning the AWS Console frontend, the backend service APIs, and the workflows that provisioned customer compute clusters.
- **UC San Diego, Halicioğlu Data Science Institute** La Jolla, California
Graduate Teaching Assistant - DSC 211: Introduction to Optimization Jan 2024 - Mar 2024
- **Amazon Web Services (AWS)** Seattle, WA
Software Development Engineer Intern (High Performance Computing group) June 2023 - Sep 2023
 - Designed internal RESTful APIs and orchestration that collapsed multi-step cluster configuration and job scheduling into single API invocations.
 - Shipped a public AWS CLI command exposing the service directly to customers.
- **UC Davis Applied Mathematics Research Group** Davis, CA
Research Assistant & Software Engineer June 2021 - June 2022
 - Developed automated reasoning pipeline that computed undiscovered Ramsey constants.
 - Applied symmetry breaking and optimization techniques to reduce the search space by 6x with preprocessing.
 - Optimized the computation to achieve up to 10x improvements in SAT clause generation.

PROJECTS

- **Minimization of CA Wildfire Impact(Web Mining, Convex Optimization, KKT, Lagrangian, Data Analysis):** Built a connectivity matrix of CA counties from humidity and fire probability, formulated wildfire prioritization as a constrained optimization problem, and solved it with integer programming over 2021 data to rank counties by financial, social and geological impact. (2023)
- **Terminal UI Board-game In Haskell(Haskell, Functional Programming, QuickCheck, Genetic Algorithm):** TUI implementation of the classic board-game Mastermind in Haskell, using monadic and functional programming, with 10 QuickCheck property tests and a genetic-algorithm AI player built on minimax. Github Link (2022)
- **Computations with Rado Numbers and Degree of Regularity(Automated Reasoning, Theory of Computation, Number Theory, Combinatorics):** Advancement in terms of Rado Numbers and the degree of regularity. Discovery of nearly 500 new mathematical constants. Results collected and formulated into research paper accepted into 2022 ISSAC conference. Co-author with Prof. Jesús De Loera and William Wesley. Github Link (2021)

PUBLICATIONS

- **Rado Numbers and SAT Computations (with J. A. De Loera and W. J. Wesley).** Proceedings of the 47th International Symposium on Symbolic and Algebraic Computation (ISSAC 2022). Pages 333-342, available online at <https://dl.acm.org/doi/10.1145/3476446.3535494>.

TECHNICAL SKILLS

- **Languages** Java, TypeScript, Bash, Rust, Python
- **Distributed Systems** REST APIs, NoSQL, Firecracker microVMs, containerization, image-based deployment
- **Cloud & Build** AWS (Lambda, ECS, SQS, DynamoDB, ElastiCache, S3, ELB), build systems, CI/CD tooling
- **Agentic AI** MCP servers, agent skills, steering docs, Claude Code, Kiro